UNDERSTAND · ANALYZE · ELEVATE

# High-Level Design

Learning Analysis Tool — analyzing what was taught vs. what was understood

## 1. Problem

Teachers deliver a session; students absorb only part of it. There is no fast, objective way to see the gap between what was **taught** and what each student **understood**. This tool closes that gap by comparing the session's concepts against the evidence students leave in the class chat.

## 2. Solution overview

The tool ingests two inputs — the lesson content and the Google Meet chat — and produces a per-student understanding report plus a class-wide view of weak spots. An LLM performs the semantic work (concept extraction and understanding judgement); deterministic Java code handles parsing, scoring, and presentation.

## 3. Architecture

A layered Spring Boot application. Each stage has one responsibility, so components are independently testable and swappable.

| Layer | Component | Responsibility |
|---|---|---|
| Input | WordExtractor (POI) | Read teacher .docx into text |
| Input | ChatParser (regex) | Group Meet chat by student |
| Intelligence | LlmClient + Prompts | Concept extraction & understanding judgement |
| Analysis | Summarizer | Coverage %, gaps, strengths, class weak spots |
| Presentation | Controller + Thymeleaf | Upload UI, dashboard, Excel export |

## 4. Data flow

1. Lesson file → WordExtractor / transcript → raw text.
2. Text → LLM → structured concept list (the ground truth).
3. Meet chat → ChatParser → per-student message bundles.
4. Concepts + each bundle → LLM → per-concept understanding level with evidence.
5. Assessments → Summarizer → coverage %, gaps, strengths, class weak spots.
6. Results → dashboard + Excel export.

### Concept data model

Each concept carries an id, name, type (definition / process / application), atomic key points (the checkable facts), and importance (core / supplementary). Student assessments reference concepts by id, keeping comparison clean.

## 5. Understanding levels

Understanding is graded on a four-point scale rather than a binary mention check — this is what makes the analysis meaningful rather than keyword matching.

| Level | Score | Meaning |
|---|---|---|
| absent | 0 | No chat evidence of the concept |
| partial | 1 | Mentioned but shallow or confused |
| understood | 2 | Explained correctly |
| mastery | 3 | Applied or questioned deeply |

Coverage % = sum of levels ÷ (concepts × 3). A concept at absent or partial is a **gap**; understood or mastery is a **strength**.

## 6. Technology stack

- **Java 21 + Spring Boot 3** — application framework, REST + server-rendered UI
- **Apache POI** — Word (.docx) reading and Excel (.xlsx) export
- **java.net.http.HttpClient** — direct LLM API calls, no SDK dependency
- **Jackson** — JSON parsing of LLM output into typed records
- **Thymeleaf** — dashboard rendering
- **Whisper** (external) — Meet audio → transcript when no Word file exists
- **JUnit 5** — test suite

## 7. Meet integration

When a Google Meet is recorded on a Workspace plan, Google saves both the recording and the chat log to Drive. The chat file (timestamp, name, message per line) is the student-evidence input. For lesson content, either a teacher's Word doc is used directly, or the recording's audio is transcribed with Whisper into the same text input. The tool runs post-session, which keeps it simple and reliable.

## 8. Design decisions & trade-offs

- **LLM for semantics, code for scoring** — keeps results explainable and testable.
- **Four-point understanding scale** — distinguishes 'mentioned' from 'understood'.
- **Chat as evidence** — zero extra student effort, but a partial signal (see limits).
- **Post-session, not real-time** — simpler, sufficient, avoids streaming complexity.
- **Human-confirmable concepts** — extraction can be reviewed before comparison.

## 9. Known limitations

Chat is a partial signal: a quiet student is not necessarily a lost one, so *absent* means 'no chat evidence', not 'no understanding'. Transcription can mishear technical terms. Both are documented and can be mitigated by pairing chat with a short post-session quiz and a domain glossary for the transcriber.